

GaussMR: Interactive Gaussian Splatting Sandbox with GPU Particles and Signed Distance Fields

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Figure 1: (a)(b) Collision with VFX particles (c) Collision with fluids (d) SDF based force fields

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1 INTERACTIVE GAUSSIAN SPLATTING FOR VR/AR/MR

Gaussian Splatting [Bernhard Kerbl and Drettakis. 2023] is a novel-view synthesis method capable of accurately capturing and rendering 3D scenes from 2D images or videos. This technique holds the potential to significantly transform the asset creation process for future games and VR/AR/MR content. To our knowledge, however, currently no real-time systems are capable of handling complex collisions and shading. Our system manipulates 3D Gaussians as GPU particles and employs a proprietary technique to construct precise Signed Distance Fields (SDF), achieving sophisticated interactions at high frame rates on Gaussian Splatting. An important consideration in XR application development is that the more realistic the environment, the more noticeable the gap becomes if the quality and accuracy of overlaid 3D objects are low. Our system addresses this challenge with its unique approach, based on real-time distance field construction for precise physical simulations and physically based rendering, enabling seamless integration into realistic spaces.

2 GPU PARTICLE-BASED GAUSSIAN SPLATTING

In our system, Gaussian Splatting is treated as GPU Particles, enabling fast and complex operations on Compute Shaders. For instance, in the scene shown in Figure 1, there are a few million Gaussians, each exhibiting real-time interactivity with individual particles of collision objects, such as those in Fluid Simulations (see Figure 1(c), 2(a)). The system incorporates a HashGrid-based spatial partitioning feature, which efficiently searches for adjacent GPU particles and rapidly detects interactions between them. This allows for advanced interactions with all Gaussians within the scene.

As a comparative example of GPU Particle-based interactive MR applications, there is Garage [Taniguchi 2021]. Garage reconstructs three-dimensional space in real-time from depth information captured by LiDAR. However, it struggled with continuous interaction in areas not captured by the sensor. Furthermore, Dimix [Taniguchi 2023] reconstructed space in real-time from depth information in 2D 360-degree image spaces to realize three-dimensional interaction, but this was limited to a 3DoF experience. Addressing these challenges, our system enables entry into realistic and interactive three-dimensional spaces based on Gaussian Splatting.



Figure 2: (a) Precise collision detection (b) Colored Gaussians as a result of collision

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3 GAUSSIAN SPLATTING WITH HIGH-QUALITY DISTANCE FIELD

A key aspect enhancing the accuracy and immersion of our system is the construction of high-speed and high-precision SDF.

3.1 SDF Noise Reduction

A major challenge when using Gaussian Splatting as the target for SDF construction is the noise component inherent in Gaussians. It is a well-known fact that Gaussian Splatting, during its training process, may inadvertently learn Gaussians representing unnecessary space, potentially leading to faint, fog-like noise in the outcome. This can become a significant artifact when included as a seed in SDF construction – an inevitable issue, given that SDF inherently seeks the distance values to the nearest seeds. To address this, we have implemented a proprietary algorithm in the SDF construction process for rapid noise elimination (see Figure 3). This algorithm effectively excludes Gaussians that could represent noise from all targeted Gaussians, while minimizing performance penalties. In our demo scene, from a few million Gaussians, it dynamically selects SDF construction targets and eliminates noise from them in about 500 microseconds. This feature, enabling the construction of smooth SDF, is a major attribute supporting our system’s high-quality interactive capabilities.

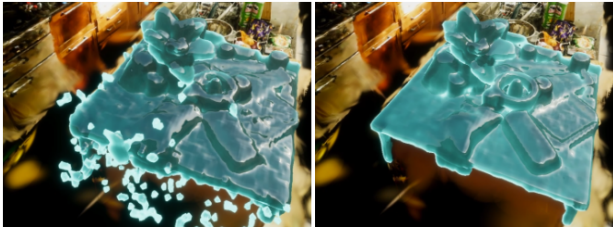


Figure 3: (a) Naive SDF (b) Noise-Eliminated SDF

3.2 Fast SDF Building

The targets for SDF construction are not limited to 3D Gaussians but include all objects appearing in the scene, such as fluid simulations. Both 3D Gaussians and fluid simulations consist of fine particles, amounting to more than 5 million vertices when rendered as quads. We have extended the Jump Flooding Algorithm (JFA) [Rong 2006] to suit Gaussian Splatting, realizing an SDF construction method appropriate for handling it. JFA builds SDF rapidly using parallel hardware like GPU by propagating the nearest seed distance values across the entire voxel grid, starting from seeds such as the vertices or centroids of the target objects. It scales well regardless of the number of seeds. However, the required parallelism and processing load increase with the resolution of the SDF, making it challenging to cover the vast scenes of Gaussian Splatting entirely. To achieve real-time processing, we have extended JFA to dynamically select target seeds and limit SDF construction to only the spaces required by the system, achieving sufficient accuracy at resolutions of 128^3 or 64^3 . Specifically, the areas of interaction with any object are dynamically selected as targets, and combined with the aforementioned noise reduction, this ensures the construction of high-quality

SDF in every frame. Furthermore, Gaussian where SDF-based interactions occur can retain the results of these interactions through the operations based on GPU Particles mentioned earlier (see Figure 2(b)). This approach allows players to experience a wide range of detailed interactions while the system maintains low overhead and high processing speed.

4 GAUSSIAN SPLATTING WITH PHYSICS SIMULATION

The fluid simulation in our system is based on SPH [Monaghan 1992] and incorporates collision simulations for interactions with SDF targeting Gaussians. Dynamic SDF-based collision detection ensures that fluids accurately follow even when Gaussians move significantly. Concurrently, all these fluid particles are utilized for advanced rendering. The use of fluid SDF in ray marching, combined with simulations of the GGX reflection model, Fresnel effects, scattering, and extinction processes, achieves a high level of realism, making it suitable for blending into Gaussian Splatting spaces based on real-world environments (see Figure 1(c),(d)). Furthermore, a smoothing process is applied in the final phase of SDF construction, resulting in very smooth surfaces.

5 XR EXPERIENCES

Our system offers an immersive XR experience, enabling players to engage in realistic environments for various activities. For example, players can sprinkle fine particles on objects to observe precise collisions, and they can also destroy scenes or splash fluids and apply colors. By recreating high-quality real-world spaces within the XR environment and enabling flexible expressions that are impossible with traditional mesh-based systems, players can experience a highly immersive Mixed Reality.

6 CONCLUSION

Our system’s ability to flexibly manipulate 3D Gaussians makes it applicable to all new emerging methods of Gaussian Splatting, continuing their impressive evolution, such as 4D Gaussian Splatting or Text to Gaussian Splatting. We believe our system presents one form of a foundational system for future immersive applications, integrating existing 3D technologies with Neural-based techniques at a high level.

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